

LA GRANDE, OR, USA

WMO: 726884

Lat: 45.283N

Lon: 118.000W

Elev: 827

StdP: 91.78

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24148

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-12.6	-9.0	-18.7	0.8	-12.4	-14.2	1.2	-8.1	15.5	3.1	14.2	1.7	2.3	320	0.608

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.9	34.1	17.3	32.6	16.9	31.1	16.4	18.5	31.5	17.7	30.7	16.8	29.3	3.6	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.7	10.8	22.5	12.5	10.0	21.7	11.4	9.3	21.0	55.4	31.3	52.6	30.6	50.0	29.3	22.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.0	10.7	9.2	-17.9	37.7	3.4	1.7	-20.4	39.0	-22.4	40.0	-24.3	41.0	-26.8	42.3		
(5)			-18.5	20.1	3.4	1.1	-20.9	20.9	-22.9	21.5	-24.8	22.1	-27.2	22.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.6	0.5	2.1	5.4	8.1	12.4	16.1	21.3	19.9	15.4	9.1	4.0	0.0
	DBStd	8.15	4.40	4.25	3.58	3.46	3.87	3.74	3.13	3.12	3.85	4.01	4.73	4.57
	HDD10.0	1330	293	222	148	77	21	2	0	0	4	66	187	311
	HDD18.3	3407	551	454	401	307	187	86	11	21	104	287	431	569
	CDD10.0	1173	0	1	6	21	95	186	349	308	165	37	6	0
	CDD18.3	209	0	0	0	0	3	19	102	70	15	0	0	0
	CDH23.3	3865	0	0	0	12	96	380	1662	1269	422	24	0	0
	CDH26.7	1703	0	0	0	1	19	126	817	583	152	4	0	0
Wind	WSAvg	3.7	5.1	4.4	3.9	3.6	3.3	3.1	2.9	2.8	2.7	3.2	4.4	5.0
Precipitation	PrecAvg	494	57	42	45	51	58	41	16	16	19	36	59	61
	PrecMax	657	86	77	86	94	154	88	58	52	58	69	136	128
	PrecMin	371	19	17	12	17	10	9	4	0	0	16	15	28
	PrecStd	77	18	16	18	22	36	20	15	12	15	16	26	23
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.2	14.2	19.1	25.0	28.9	33.6	37.4	36.3	32.9	27.0	18.0	12.2
		MCWB	6.0	7.8	9.7	13.2	15.6	18.3	18.5	17.5	16.2	13.8	9.9	7.2
	2%	DB	8.0	12.0	17.2	22.1	26.2	30.0	35.0	33.9	31.0	22.4	16.0	9.0
		MCWB	4.4	6.7	8.8	12.1	14.8	16.6	17.6	16.9	15.6	11.6	8.7	5.6
	5%	DB	7.3	9.5	14.1	18.5	23.1	27.6	32.9	32.4	27.9	19.1	12.9	7.4
		MCWB	3.8	5.4	7.2	10.4	13.5	15.6	17.2	16.5	14.5	10.4	7.3	4.2
	10%	DB	6.2	7.8	12.4	16.0	21.1	25.2	31.6	30.2	25.9	17.2	11.3	6.1
		MCWB	2.8	4.1	6.7	8.7	12.4	14.5	16.9	15.8	13.7	9.4	6.4	2.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.7	8.7	10.7	14.1	17.2	19.3	20.1	18.9	17.8	14.3	11.2	7.4
		MCDB	9.1	13.5	17.6	22.6	27.0	31.3	32.5	32.9	29.9	25.2	16.2	11.0
	2%	WB	5.1	6.9	9.2	12.3	15.5	17.5	18.9	17.8	16.4	12.4	9.4	5.7
		MCDB	7.7	11.0	15.7	20.6	24.3	28.1	31.8	31.7	28.4	20.8	15.0	8.5
	5%	WB	4.1	5.6	8.0	10.6	14.1	16.1	18.0	17.0	15.2	11.0	7.8	4.5
		MCDB	6.8	9.1	13.8	17.4	22.3	26.3	31.2	30.5	26.4	18.3	13.0	7.2
	10%	WB	3.1	4.6	6.9	9.3	12.9	15.0	17.2	16.1	14.0	9.8	6.6	3.2
		MCDB	5.8	7.7	11.9	15.2	19.8	24.4	30.3	29.0	24.7	16.3	11.0	5.8
Mean Daily Temperature Range	5% DB	MDBR	6.2	8.0	11.0	12.3	13.1	14.5	18.9	19.3	17.9	13.8	8.7	6.3
		MCDBR	7.0	10.0	14.9	18.6	18.2	19.2	21.8	23.0	23.0	19.0	12.0	6.7
		MCWBR	4.9	6.5	8.9	10.5	9.4	9.3	9.2	9.7	11.0	10.4	7.3	4.8
	5% WB	MCDBR	6.6	9.3	13.6	16.9	16.3	17.8	20.1	20.8	20.6	17.0	11.0	6.5
		MCWBR	4.9	6.1	8.4	9.8	8.5	8.8	8.9	8.9	9.9	9.6	6.9	4.8
Clear-Sky Solar Irradiance	taub	0.297	0.291	0.308	0.369	0.374	0.358	0.359	0.363	0.350	0.315	0.295	0.282	
	taud	2.375	2.465	2.485	2.297	2.328	2.405	2.422	2.403	2.417	2.521	2.513	2.462	
	Ebn at Noon	812	902	930	890	894	908	902	887	869	861	819	800	
	Edh at Noon	75	82	93	123	123	115	111	109	99	78	65	62	
All-Sky Solar Radiation	RadAvg	1.39	2.30	3.48	5.00	6.03	7.02	7.65	6.50	4.83	3.00	1.64	1.10	
	RadStd	0.16	0.27	0.35	0.39	0.52	0.43	0.30	0.36	0.33	0.24	0.13	0.14	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (14 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page